Evaluating the Contribution of Road Network Topology to Surface Urban Heat Island Estimation in Davao City Using Graph Neural Networks

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CHAPTER I
INTRODUCTION

Urbanization and changes in land use have intensified environmental challenges in cities facing the combined pressures of development and climate change. As urban populations grow, more residents are exposed to elevated temperatures, increasing the need for locally appropriate heat-mitigation measures. A global assessment of future urban temperatures emphasizes the potential role of nature-based solutions in limiting these impacts [1]. Satellite evidence also shows that urban surface warming can exceed warming in surrounding areas: an analysis of more than 2,000 city clusters found that the mean surface warming trend in urban cores was 29% greater than in rural backgrounds during 2002-2021 [2]. The combined influence of urban development and climate change has been investigated through complementary methods. A study of Lagos separated urbanization-related and broader climatic contributions to changes in urban surface temperature, demonstrating the importance of distinguishing local land-cover effects from background warming [3]. Scenario-based research in Tehran also examined urbanization and climate change in relation to urban heat and thermal comfort [4].

At the global scale, projections integrating future urban expansion and climate change indicate that urbanization can add to local surface warming through 2100 [5]. Land surface temperature (LST) describes the thermal condition of the ground, vegetation, roofs, and other surfaces as estimated from thermal infrared observations [6]. It differs from the temperature of the overlying air. A surface urban heat island (SUHI) occurs when urban surfaces are warmer than a specified surrounding non-urban background; SUHI intensity expresses this difference rather than the absolute surface temperature [7]. Consequently, high LST alone does not establish the magnitude of SUHI intensity. The reference surface must be defined explicitly because differences in rural land cover can change the resulting urban-rural temperature contrast [8]. Research comparing crowdsourced air

temperatures with satellite observations further shows that surface and air-temperature heat islands can differ in magnitude and their relationships with environmental characteristics [9]. The relevance of urban thermal information extends to adaptation planning. A review of resilience to weather and climate extremes identifies infrastructure, socioeconomic conditions, and governance as interacting influences on urban adaptive capacity [10]. Research on urban heat-risk management similarly emphasizes differentiated vulnerability, participation, uncertainty, and the sequencing of adaptation actions [11]. Consequently, identifying warmer urban surfaces is a useful component of environmental assessment, but deciding where and how to intervene requires evidence about local conditions and the people affected. This distinction helps define the contribution of a surface-temperature model within a broader planning process.

The intensification of urban heat is particularly critical in Southeast Asia, where high humidity and accelerating development expose growing populations to severe thermal risks [12], [13]. Regional projections indicate that the combined pressures of climate change and urban expansion will significantly elevate local surface temperatures [14], [15]. While extensive research has characterized broad surface thermal variations across tropical cities [16],[17], there remains a critical need for high-resolution spatial modeling to inform localized urban planning. Consequently, accurate, street-level intra-urban heat mapping serves as an essential first step before communities can evaluate and implement practical cooling interventions [18], [19].

In the Philippines, long-term meteorological records provide evidence of changes in temperature extremes. A national assessment analyzed daily precipitation and near-surface air temperature for 1951-2010 [20]. A subsequent analysis of observations from 34 meteorological stations for 1951-2018 identified differences in temperature trends between urban and rural stations, with stronger warming in indices associated with minimum temperature [21]. These atmospheric observations establish a national context, but they should not be treated as direct

measurements of satellite-derived SUHI intensity or as proof that every observed temperature change is caused by urbanization. Philippine remote-sensing studies provide more direct evidence of surface thermal variation. An analysis of Metropolitan Manila found that increasing built-up indicators and declining vegetation indicators were associated with higher LST in Metropolitan Manila between 2001 and 2019 [22]. A study of Manila City combined satellite and meteorological information to assess intra-urban thermal variation in Manila City and locate areas for further intervention assessment [23]. Beyond the capital region, research examined land-cover changes in the Cagayan de Oro River basin from 2003 to 2020 and reported higher mean LST over built-up areas and bare land than over forest, mangroves, and water bodies [24]. In fact, a localized heat-risk assessment in Quezon City integrates environmental exposure and socioeconomic vulnerability in a weighted index, illustrating how densely developed and vegetation-poor areas can overlap with socially vulnerable communities [25].

Atmospheric modeling offers complementary evidence about the Philippine urban environment. A numerical simulation study used the Weather Research and Forecasting model coupled with an urban canopy model to examine urban land cover and anthropogenic heat during extreme heat events in Metro Manila [26]. A related study evaluated urban canopy model configurations for air temperature and relative humidity during a high-heat event, emphasizing the importance of appropriate urban morphology information [27]. These studies concern atmospheric processes and do not establish that heat travels along road-network connections. At the national scale, a heat-health assessment combined MODIS-derived thermal information with population data for a heat-health assessment [28].

These thermal pressures are also evident in Davao City. A local study mapped the occurrence of hot and cold surface-temperature locations from 1994 to 2019 and examined associations with vegetation, built-up indicators, albedo, elevation, and population density. Its quantile-regression results identified the Normalized Difference Built-up Index (NDBI) as the most significant predictor at the 50th, 75th, and 90th LST quantiles, while other predictors were more important

at lower quantiles [29]. This variation cautions against assuming that a single environmental factor explains thermal conditions throughout the city. Another local investigation reported higher surface temperatures in densely built-up, vegetation-deficient parts of Davao than in greener areas [30]. These local studies establish a basis for further surface-temperature modeling, but they do not determine whether road-network connectivity adds predictive information beyond the environmental attributes observed at each location. Additional Landsat-based research in Davao City for 2018-2022 reports a positive association between NDBI and LST and a negative association between NDVI and LST [31].

Satellite remote sensing provides spatially extensive information for examining these local differences, subject to cloud cover, retrieval uncertainty, and the spatial and temporal characteristics of the observations [6], [7]. Machine-learning approaches used in urban heat research include Random Forest (RF), convolutional neural networks (CNNs), and other neural architectures [32]. A further comprehensive review examines the range of machine-learning methods used for urban heat-island assessment [33]. However, predictive performance depends on how the data, spatial relationships, and evaluation procedure are represented. A review of environmental spatial machine learning identifies several ways to account for spatial autocorrelation, including spatial covariates and training-test partitioning, and finds no single approach that is appropriate for every application [34].

Empirical comparisons provide a reason to retain a conventional benchmark. Fine-scale urban heat modeling has compared Random Forest with multiple linear regression using air temperature and a broad set of environmental and urban predictors [35]. Although the response variable differs from satellite LST, the study demonstrates the relevance of evaluating nonlinear models against alternatives under a defined task. An open-source UHI-mapping framework for Lagos uses a U-Net architecture with satellite and urban geospatial inputs [36], illustrating a raster-based approach to a related thermal problem.

Although Random Forest can capture nonlinear relationships between environmental predictors and LST, a conventional tabular implementation does not explicitly aggregate information through road-network connections unless such information is encoded in its inputs. Standard raster CNNs represent local spatial context through grid neighborhoods; those neighborhoods do not explicitly distinguish locations connected by a road from locations that are merely close in geographic space. A road graph provides a different representation: intersections and intermediate road points become nodes, and connecting road segments define their adjacency. Street-graph research demonstrates the usefulness of network-based units for characterizing urban morphology, although that work uses geometric information beyond connectivity alone [37]. In this study, road adjacency is treated as a candidate relational representation whose predictive contribution must be tested. It is not assumed to be a physical heat-transfer pathway or a complete description of street-canyon geometry.

A graph representation also requires a clear interpretation of its edges. Research on regional surface heat islands has combined morphological spatial pattern analysis and graph-based connectivity measures to evaluate the configuration of thermal patches [38]. Such a thermal-landscape graph differs from a street graph because its connections are defined for a different analytical purpose. This distinction matters when adapting graph methods: an adjacency matrix is a modeling choice, and evidence obtained with thermal patches cannot by itself validate road adjacency. In the proposed model, connected road locations provide neighborhoods for information aggregation, with their usefulness assessed through prediction errors.

Graph neural networks (GNNs) offer mechanisms for learning from both node attributes and the relationships specified by a graph [39]. Environmental applications provide relevant methodological precedents. One study developed graph-based spatiotemporal interpolation using a Voronoi-adjacency representation [40]. Another study applied a GCN framework to daytime and nighttime SUHI modeling in Hong Kong, integrating built and natural urban features [41][45]. A recent review examined graph construction and spatial-temporal learning across

several urban environmental applications [42][46]. These studies support investigating graph-based thermal estimation, but results obtained with other graph definitions, environmental variables, and locations cannot establish the benefit of Davao City's road topology in advance.

While recent environmental applications have successfully utilized Graph Neural Networks (GNNs) for tasks such as spatiotemporal interpolation and road-surface temperature forecasting [40], [42], [43], these models are primarily designed for short-term atmospheric forecasting. Estimating annual satellite-derived LST requires a different temporal approach. Consequently, while these graph architectures provide a methodological precedent, their results motivate further investigation without justifying claims about subpixel road temperatures or physical heat movement in the present study.

The present study will represent road-associated locations in Davao City using an OpenStreetMap (OSM) network acquired and processed with OSMnx, which provides methods for constructing and analyzing street graphs [44]. Nodes will include intersections and intermediate points along road segments. Each node will be associated with historical LST, the Normalized Difference Vegetation Index (NDVI), and NDBI. NDVI and NDBI will be treated as spectral indicators of vegetation and built-up characteristics, respectively; NDBI will not be interpreted as a direct measurement of impervious-surface density. Road connectivity will allow the model to combine information from connected locations. The hypothesis is that this relational information may contribute to estimation after accounting for the historical environmental features at each node. Combining OSM with satellite and environmental information has also been explored in a multi-domain urban-planning framework that addresses land use, building footprints, traffic, and air quality [45]. This provides a precedent for integrating volunteered geographic information with remotely sensed inputs, while the present study has a narrower thermal-estimation purpose. OSM coverage and the alignment of road geometry with satellite observations will consequently be documented as properties of the

input data. A more detailed road graph will not be treated as evidence of equally detailed thermal measurements.

The proposed architecture combines a Long Short-Term Memory (LSTM) encoder with GraphSAGE neighborhood-aggregation layers. LSTM provides the temporal sequence-learning component [46], while GraphSAGE provides a method for learning node representations by aggregating neighborhood information [47]. In this chapter, the specific model is called LSTM-GraphSAGE to distinguish its architecture from the broader GNN family. Annual median environmental features from a continuous six-year historical sequence (2019–2024) will be used to estimate the 2025 LST associated with each node. SUHI intensity will then be derived for urban nodes by subtracting a common, independently specified non-urban reference LST for 2025. This 7-year historical target-year experiment evaluates spatial estimation using a robust multi-year sequence; it does not establish operational short-term forecasting or reliable prediction under future infrastructure changes.

The methodological gap is whether explicitly representing road-network connectivity improves estimation beyond what can be learned from the same historical satellite-derived features without information exchange between locations. Comparing LSTM–GraphSAGE with Random Forest alone would not isolate this contribution because both the architecture and the use of adjacency would change. The study will therefore compare the road-connected model with a matched configuration in which all edges between distinct nodes are removed, and only self-loops are retained. In this configuration, the LSTM encoder and GraphSAGE layers remain, but each node processes only its own historical features and representation, without receiving information from other nodes. The resulting model operates as a shared node-wise predictor at the same road-associated locations. Both neural configurations will use the same input features, layer dimensions, and training procedure. A Random Forest model using the same historical predictors will provide a separate conventional benchmark. All configurations will be assessed on the same held-out observations.

Through these comparisons, the study will evaluate both overall estimation performance and the added predictive value of the chosen road-network representation. Its outputs will include estimated LST, derived SUHI intensity at urban road-network nodes, and maps of locations with elevated estimated SUHI intensity. These outputs are intended to support the interpretation of surface thermal variation within the defined urban study area of Davao City. Neither improved performance nor the ability to identify hotspots will be assumed before evaluation.

Research Questions

This study seeks to evaluate whether road-network topology contributes useful information to the estimation and mapping of satellite-derived SUHI intensity in Davao City. Specifically, it will address the following questions:

1. How accurately can the proposed LSTM–GraphSAGE model, integrating OSM road-network topology and historical satellite-derived environmental features, estimate 2025 land surface temperature (LST) at urban road-network nodes in Davao City?
2. What change in estimation performance results from incorporating road-network connectivity compared with (a) a matched configuration where inter-node connections are disabled, and (b) a spatial-proximity graph (e.g., K-Nearest Neighbors) that connects nodes based purely on geographic distance?
3. How does the predictive performance of the proposed model compare with a Random Forest regression baseline using the same historical predictors and held-out observations, based on root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2)?

Significance of the Study

This study will provide spatially explicit information on SUHI intensity and an empirical assessment of road-network topology as a representation for thermal estimation. Its potential contributions concern urban planning, local environmental assessment, and reproducible geospatial machine learning.

GeoAI and Computer Science. The comparison between road-connected and connectivity-disabled configurations will examine the predictive value of an explicit network representation while controlling the principal architectural differences. The separate Random Forest benchmark will assess the complete model against a conventional machine-learning approach. A finding that road connectivity provides little or no improvement would also be informative because it would clarify the limits of this representation for the selected data and task.

Remote Sensing and Urban Climate Research. The study will relate satellite-derived thermal and spectral information to road-associated locations while maintaining the distinction between LST and reference-based SUHI intensity. Documenting the reference-area definition, imagery coverage, and spatial support of the estimates will help make the results interpretable and comparable with subsequent local studies.

Urban Planning and Engineering. The resulting maps may help the City Planning and Development Office and the City Engineer's Office identify road-associated locations with elevated estimated SUHI intensity for further site assessment. Such information may inform the consideration of shading, vegetation, and surface-material interventions alongside field observations and engineering requirements. The study will not determine the optimal intervention or demonstrate the cooling effectiveness of cool pavements, permeable materials, or urban greening.

Local Government and Environmental Management. The study may provide supporting spatial evidence for discussions of land-use planning, zoning, and urban greening priorities. The maps can indicate where further environmental investigation is warranted, but they will not constitute a heat-health warning

system. Decisions involving disaster response or public health would require additional meteorological, exposure, and vulnerability information.

Open and Reproducible Geospatial Modeling. The use of OSM, Landsat data, and documented processing procedures will support reproducibility. Recording acquisition dates, quality-screening rules, graph-construction decisions, model settings, data partitions, and evaluation procedures will enable other researchers to examine the workflow and assess its suitability for comparable urban contexts.

Scope and Delimitations

The geographic extent of road-network acquisition will follow the Davao City administrative boundary obtained from OSM. This acquisition extent includes urban, agricultural, rural, and mountainous areas; however, citywide acquisition does not mean that every road node represents an urban SUHI observation. The principal estimation and interpretation will focus on nodes within the continuously built-up urban area, which will be operationally defined using an objective, reproducible land-cover classification dataset (e.g., ESA WorldCover) to eliminate manual boundary bias. Urban evaluation locations and non-urban reference surfaces will be distinguished through documented geographic and land-cover masks. Areas will not be designated urban, excluded, or retained solely to obtain a more connected graph. The exact masks, road-type filters, and treatment of disconnected components will be specified in the methodology and reported with the resulting coverage.

The study will use Landsat 8 Collection 2 Level-2 surface reflectance and surface temperature products for 2019-2025 to extract surface infrastructure indices, vegetation indices, and annual median thermal observations. The USGS surface temperature product provides the thermal observations, while NDVI and NDBI will be computed from the corresponding reflectance data using the appropriate product scaling and quality information [48]. Cloud, cloud-shadow, and invalid observations will be screened before constructing annual median composites. Historical LST, NDVI, and NDBI from 2019 to 2024 will form the

sequential model inputs, and the 2025 LST composite will provide the target observations... To preserve a mathematically consistent 16-day orbital sampling frequency across all seven years and prevent artificial warming biases, multisensor fusion with Landsat 9 is explicitly excluded from the analysis.

The model will estimate LST, from which urban-node SUHI intensity will be derived. For an urban node i and target year t, the operational relationship is SUHI(i,t) = LST(i,t) - LSTref(t), where LSTref(t) is the median of valid annual-composite LST pixels within the designated non-urban reference mask for the same year. The estimated SUHI will use the model-estimated urban LST and the same reference value used for the satellite-derived comparison. The reference mask will be defined independently of model prediction errors and will retain non-urban reference pixels even when they are outside the modeled urban graph. Its land-cover composition and elevation comparability with the urban area will be documented, and water surfaces will not be pooled into the terrestrial rural reference. Because reference selection affects SUHI interpretation, the reference boundary and eligibility criteria must be fixed before final evaluation [8]. This procedure produces a reference-based surface-temperature contrast, not an independently measured air-temperature heat island.

The road graph will contain intersections and intermediate points along retained road segments. Node densification changes the locations at which features are assigned and predictions are reported; it does not create finer independent thermal observations. Spatial detail remains constrained by the native and processed support of the Landsat observations. Accordingly, references to the street-network level describe the placement of estimates on a road graph, not direct measurement of pavement temperature or pedestrian-level microclimate. The road-network representation will also depend on OSM completeness, positional accuracy, attribute quality, and the date of extraction. An OSM network snapshot will not be assumed to reproduce all road changes occurring during 2019-2025.

The comparative analysis will evaluate four configurations: (1) LSTM–GraphSAGE with OSM road adjacency, (2) a matched configuration

without information exchange between distinct nodes, (3) a matched configuration utilizing a spatial-proximity graph (e.g., K-Nearest Neighbors) based on geographic distance, and (4) Random Forest regression. To ensure the matched configurations isolate the contributions of road and spatial adjacency without altering the underlying neural architecture, 'without information exchange' is defined computationally by restricting the adjacency matrix exclusively to self-loops (i.e. aggregating only self-information). These comparisons will assess the contribution of road-network connectivity and the performance of the proposed approach relative to a conventional machine-learning benchmark. Evaluation will account for spatial dependence and shared satellite observation support [38]. The model configurations, training procedures, and spatial train/validation/test partitioning strategy will be specified in Chapter 3.

Predictive performance will be evaluated for LST using RMSE, MAE, and R^2. SUHI intensity will be derived from estimated LST using a designated target-year non-urban reference and used primarily for spatial interpretation and mapping. It will not be treated as an independently predicted target or subjected to a separate, redundant assessment of prediction error. Maps of estimated LST and derived SUHI intensity will present their spatial distribution at urban road-network nodes rather than constitute a separately evaluated hotspot-identification system.

The study will use satellite observations covering 2019–2025, with conclusions restricted to the input periods and target years evaluated. Annual aggregation does not support conclusions about seasonal behavior, diurnal thermal patterns, or short-term heat-event forecasting. The analysis excludes ground-based air-temperature prediction, thermal comfort, heat index, human heat exposure, and health-risk estimation. It will not simulate physical heat transport along roads, quantify the causal effects of individual road connections, or predict the cooling outcomes of infrastructure changes. Its conclusions will be limited to comparative LST estimation performance and mapped surface thermal conditions under the documented observation and SUHI reference definitions.

CHAPTER II
REVIEW OF RELATED LITERATURE

This chapter reviews the literature and implemented systems relevant to evaluating the contribution of road network topology to surface urban heat island estimation in Davao City. It is organized into two main sections.

Related Literature discusses thermal concepts, Landsat-based environmental observations, research in tropical and Philippine cities, road networks and spatial graph representations, machine-learning approaches, and evaluation methods, concluding with a synthesis of the research gap.

Related Systems examines satellite-based mapping, conventional machine learning, raster-based deep learning, graph-based estimation, and spatiotemporal temperature forecasting systems, followed by a comparative analysis and the positioning of the proposed system.

2.1 Related Literature

This section establishes the conceptual and methodological foundations of the study. It begins with land surface temperature and surface urban heat islands, followed by Landsat-based environmental observations and urban surface-temperature research in tropical and Philippine cities. It then discusses road networks and spatial graph representations, machine learning for annual LST estimation, and methods for evaluating spatial prediction and the contribution of road connectivity. Finally, the section synthesizes the reviewed literature to identify established findings, unresolved questions, and the research gap addressed by the proposed study.

2.1.1 Land Surface Temperature and Surface Urban Heat Islands

Understanding urban thermal dynamics requires a clear physical definition of the target variable. Land Surface Temperature (LST) is physically defined as the

radiometric skin temperature, representing the radiation emitted from the top few millimeters of the Earth's surface [49]. Extracting this measurement from space-based thermal infrared (TIR) radiance is inherently an "ill-posed" inverse problem, as it requires simultaneous estimation of atmospheric corrections and surface emissivity [49]. Consequently, satellite-derived LST establishes the thermal condition of the physical ground surface, distinguishing it fundamentally from ambient canopy air temperature. Because urban surfaces naturally absorb and retain heat differently than their surroundings, high absolute LST alone does not constitute an urban heat island. Instead, the Surface Urban Heat Island (SUHI) represents the radiative temperature difference between urban infrastructure and neighboring non-urban surfaces [7]. To properly evaluate this phenomenon, it is necessary to derive a differential intensity metric ($\Delta LST = LST_{urban} - LST_{non-urban}$). Empirical evidence from over 2,000 global city clusters demonstrates the necessity of this contrast, revealing that urban cores experience surface warming rates 29% faster than their surrounding rural backgrounds [2].

However, SUHI intensity is highly sensitive to the geographic and methodological definition of the non-urban reference. Systematic examinations across hundreds of global cities confirm that significant variations in SUHI intensity are often driven by the vegetation composition of the rural baseline, rather than the urban core itself; fixing the reference land cover stabilizes the metric [8]. Furthermore, large cities frequently "bleed" heat outwards, exhibiting widespread LST increases that extend into peri-urban and rural areas [17]. To isolate the thermal impact of urban infrastructure from regional macro-climatic trends—which can account for a significant portion of long-term surface warming [3]—the non-urban reference buffer must be carefully defined and placed far enough from the urban edge to prevent contamination from peri-urban thermal expansion.

Finally, while LST provides unparalleled spatial coverage for evaluating

the built environment, it must not be conflated with pedestrian heat exposure. Empirical comparisons between satellites and tens of thousands of crowdsourced weather stations reveal that satellite LST can produce a sixfold overestimate of UHI intensity compared to canopy air measurements during the daytime [9]. Furthermore, city-level satellite analyses often fail to capture socially salient microhotspots, behavioral adaptations, and lived experiences of heat exposure that are identifiable through participatory community mapping [19]. Therefore, while satellite LST is an optimal metric for evaluating the surface thermal properties of urban infrastructure, it cannot be directly interpreted as an assessment of human health risk or ambient thermal comfort.

2.1.2 Landsat-Based Environmental Observations

Accurate estimation of urban thermal dynamics relies on robust, validated satellite data. This study utilizes the Landsat 8 Collection 2 Level-2 (C2L2) Land Surface Temperature product, which has demonstrated strong overall reliability ($R^2 = 0.91$, $r = 0.98$) in comprehensive ground validations [50]. However, it is critical to recognize the sensor's environmental constraints. Validation studies indicate that the C2L2 LST product exhibits a systematic warm bias of +1.75°C and experiences significant retrieval errors when surface temperatures exceed 40°C or when atmospheric water vapor is high [50]. Acknowledging these limitations validates the primary data source while directly establishing the expected sensor biases when operating in humid, high-temperature tropical environments like Davao City. Processing massive volumes of multi-year satellite imagery is computationally impractical using standard desktop resources [51]. Consequently, this study relies on Google Earth Engine (GEE) to access radiometrically calibrated Tier 1 Landsat data, which is specifically suited for time-series analysis [52]. To ensure data quality, pixel-level quality assessment (BQA) bands are applied to rigorously mask clouds, cloud shadows, and cirrus artifacts. Rather than relying on potentially noisy single-date scenes, a temporal median value is computed for each band across the year [53]. This operational workflow

statistically filters out unmasked transient cloud artifacts and sensor dropouts, defending the use of annual median composites (2019–2025) to ensure consistent comparability across the study period.

To characterize the urban landscape, the model incorporates spectral indicators such as the Normalized Difference Built-up Index (NDBI) and the Normalized Difference Vegetation Index (NDVI). NDBI is mathematically designed to automate the mapping of built-up areas by exploiting the physical principle that urban surfaces reflect shortwave infrared (SWIR) light more strongly than near-infrared (NIR) light [54]. Conversely, NDVI maintains a strong, well-documented negative correlation with LST across multiple spatial resolutions [55]. Together, these indices justify the use of baseline environmental features in thermal regression. However, this study scopes NDBI strictly as an automated spectral proxy for urban structures, not as a direct physical measurement of impervious surface density, given its known limitations in isolating complex land covers.

Finally, the spatial resolution of the Landsat sensor introduces inherent limitations when mapping highly localized urban features. While Landsat provides a 30-meter spatial resolution, thermal infrared data is often restricted by instrumental limitations that cause a "blending effect," or mixed pixel [56]. A single 30-meter thermal footprint inevitably aggregates the radiant emissions of asphalt roads, adjacent buildings, and roadside trees into a single value, making it difficult to isolate specific urban characteristics. This blending effect explicitly defends the study's delimitations: while assigning satellite estimates to densified road nodes allows for the evaluation of the network's topological thermal patterns, it does not create true micro-scale pavement or street-canyon measurements.

2.1.3 Urban Surface-Temperature Research in Tropical and Philippine Cities

The intensification of urban heat is particularly severe in Southeast Asia, where rapid urbanization intersects with a humid equatorial climate. Multi-decadal satellite assessments across Southeast Asian capitals reveal significant and persistent land surface warming trends (e.g., 0.215°C to 0.350°C per year), driven primarily by the extensive loss of vegetation and the proliferation of impervious surfaces [14]. Comparative studies of dynamic metropolises like Bangkok and Ho Chi Minh City demonstrate that these landscape shifts weaken natural cooling mechanisms and dramatically alter surface energy balances [16]. Furthermore, the thermal impact of these megacities is not confined to the urban core; larger cities exhibit an outward "bleed" of heat, propagating intense LST increases into surrounding peri-urban and rural areas [17]. This establishes the macro regional context for the present study, demonstrating that environmental thermal pressures are intensely magnified in tropical climates that are highly sensitive to land-cover perturbations.

These regional thermal pressures are clearly manifested within the Philippines. At the national atmospheric level, meteorological records (1951–2018) indicate that urbanization significantly accelerates minimum nighttime temperatures over daytime temperatures, shrinking the diurnal temperature range with the strongest urban-rural differences occurring during the cool dry season [21]. In terms of surface heat, satellite time-series studies in Metropolitan Manila document severe LST increases (e.g., 4.02°C from 2001 to 2019) that are directly correlated with rising NDBI and declining NDVI [22]. Furthermore, spatial analyses in Quezon City confirm that these thermal variations are highly localized and non-random, forming clustered intra-urban heat islets that overlap strongly with densely urbanized, vegetation-poor zones [25]. These studies validate the predictive link between built-up indices and elevated LST in the Philippines, while reiterating that spatially clustered surface heat behaves differently than general atmospheric warming.

Beyond the capital region, localized studies confirm that these surface

thermal dynamics are highly active in Mindanao. Research in the Cagayan de Oro River basin observed a mean LST increase of 5.48°C over 17 years as croplands and built-up areas expanded, with built surfaces consistently recording the highest temperatures [24]. These patterns hold true within Davao City. Spatiotemporal analyses across the city's congressional districts reveal peak surface temperatures approaching or exceeding 38°C in highly urbanized barangays [30]. Quantile regression models (1994–2019) identify NDBI as the single most significant predictor of high-quantile LSTs in Davao's core districts [29]. More recent Landsat 8 analyses (2018–2022) confirm strong spatial correlations between NDBI ($r = 0.6216$) and LST, explicitly highlighting major arterial corridors as zones of high thermal concentration [31].

While these local studies successfully establish that built infrastructure elevates surface temperatures across Davao City, they treat locations as isolated pixels or administrative polygons using global regression models. However, recent comparative spatial modeling in Metro Manila reveals that extreme heat clusters closely follow major transport arteries, with primary roads exhibiting a massive, highly significant spatial association with heat propagation (coefficient = 0.540) [19]. The presence of significant spatial lag parameters indicates that heat in one grid cell is strongly influenced by neighboring cells—evidence of spatial spillover at neighborhood scales. Despite this evidence, no local studies have investigated whether the explicit network topology and structural connectivity of the road system itself contribute useful predictive information for estimating surface temperatures along urban transit infrastructure. This unresolved methodological gap directly motivates the proposed graph neural network model, which tests road-network connectivity against traditional, isolated spatial estimation.

2.1.4 Road Networks and Spatial Graph Representations

Road networks can be modeled as spatial graphs, with intersections represented as nodes and connecting street segments as edges in a primal

representation [57]. However, road connectivity concerns the connections represented by these edges, whereas geographic proximity concerns spatial distance between locations [58]Moreover, topology describes connections and relative configuration, while geometry describes spatial position and shape, including lengths and angles [59]. These distinctions provide a conceptual basis for examining road adjacency as a candidate relational representation for LST estimation; they do not establish its predictive contribution.

Building on these graph concepts, OpenStreetMap (OSM) provides collaboratively produced geographic data for representing road networks [60]. OSMnx is a Python package that supports the acquisition, construction, and analysis of street-network models from OSM [61]. A global study demonstrated a reproducible workflow for modeling urban street networks using OSM and OSMnx, providing a methodological foundation for consistent network representation [62]. Furthermore, research in Jakarta, Indonesia, combined OSMnx-based network analysis with history-based OSM quality assessment, illustrating the importance of examining the underlying data when interpreting network models [63]. Together, these studies support constructing an OSM-derived road graph while considering its representation and data quality; its predictive contribution to LST estimation remains a separate empirical question [last two citations mentioned

The literature on road-network modeling distinguishes intersections from intermediate points used to describe road geometry. In fact, OSM-derived networks may contain vertices that represent road curvature rather than intersections, while complex intersections may be represented by multiple nodes [59]. Accordingly, graph simplification and node consolidation address different aspects of how mapped features correspond to real-world intersections and street segments [59]. These distinctions inform the present study’s use of intersections and intermediate road points as road-associated observation locations, while leaving the predictive value of this representation for LST estimation to empirical evaluation [58].

Beyond road connectivity, graph-learning literature represents relationships and dependencies among entities through different graph structures [64]. In environmental prediction, an air-quality study distinguished spatial proximity, functional similarity, and temporal-pattern similarity among monitoring stations. If anything, spatial proximity was represented using geographic distance, whereas functional similarity considered surrounding points of interest, demonstrating that graph connections can encode different kinds of relationships [65]. In contrast, primal road graphs represent connections through street segments, which describe a different relation from geographic closeness or similarity in surrounding characteristics [58], [65]. Together, these approaches provide a conceptual basis for comparing road-connected and spatial-proximity configurations for LST estimation, while leaving their relative predictive value to empirical evaluation.

Nevertheless, the suitability of an OSM-derived road graph also depends on the quality and temporal coverage of its underlying data [66], [67]. A review of volunteered geographic information identifies several dimensions of quality, indicating that completeness alone is insufficient to characterize a dataset's suitability [66]. Similarly, a global assessment of OSM roads found geographic variation in completeness, highlighting the need to consider local conditions [67]. History-based assessment can examine recorded changes and data currency, as illustrated by research on OSM object histories and the Jakarta network study [63]. However, the OSM history literature concerns changes recorded in the database; interpreting these records as a complete chronology of real-world road development requires caution [68]. These findings inform the use of an OSM-derived graph alongside historical environmental observations, while leaving its correspondence to road conditions throughout the study period as a limitation to assess [63], [68].

2.1.5 Machine Learning for Annual LST Estimation

Land Surface Temperature (LST) is physically defined as the radiometric skin temperature emitted from the outermost layer of terrestrial surfaces, representing direct directional radiance rather than the convective thermodynamic properties of the atmospheric boundary layer [6]. In satellite remote sensing, retrieving LST from spaceborne thermal infrared (TIR) sensors is an inherently ill-posed inverse problem that requires simultaneously decomposing observed radiance into land surface emissivity and coupled atmospheric components [6], [7]. Because instantaneous satellite scenes are intermittently obscured by monsoonal cloud formations, cloud shadows, and episodic rainfall in tropical environments such as Davao City, constructing annual composite products—specifically annual median LST from cloud-screened Landsat observations—serves as an indispensable pre-processing strategy to suppress high-frequency weather noise, eliminate seasonal cloud voids, and isolate underlying climatological conditions [7], [33].

Global empirical investigations demonstrate that urban land surface warming outpaces surrounding rural warming, driven primarily by the replacement of natural vegetative surfaces with high-admittance impervious materials [2]. In a multi-decadal analysis of satellite LST across more than 2,000 city clusters, researchers observed that urban cores exhibited an average surface warming rate of 0.50 ± 0.20 K decade^{-1}, outpacing rural background areas by roughly 29%, with direct urban land expansion contributing more than 0.23 K decade^{-1} of the warming trend across developing Asian metropolitan areas [2]. In the Philippines, regional landscape dynamics similarly induce pronounced thermal shifts; across the Cagayan de Oro River basin in northern Mindanao, an empirical increase of 5.48°C in mean LST was recorded between 2003 and 2020, with built-up areas and barren land exhibiting the highest surface thermal values while croplands and forest cover maintained the lowest [24]. These multiscale findings establish that radiometric surface temperatures are fundamentally modulated by land-cover

transitions and built infrastructure, confirming the theoretical necessity of modeling LST as a function of biophysical and anthropogenic surface indices.

At the municipal scale in Davao City, localized empirical research has established strong, persistent relationships between satellite-derived thermal patterns and remote-sensing surface indices. Evaluating Davao City's urban thermal environment between 1994 and 2019, an empirical investigation mapped spatiotemporal hot and cold spots using Landsat 8 and Synthetic Aperture Radar data within multiple linear and quantile regression frameworks [29]. Their quantile modeling revealed that the Normalized Difference Built-up Index (NDBI) served as the primary explanatory driver of elevated surface temperatures across the median and upper thermal quantiles (50th, 75th, and 90th percentiles), whereas surface albedo and digital elevation dominated at lower quantiles (10th and 25th percentiles), and atmospheric particulate matter (PM_{10}) demonstrated no statistically significant explanatory power [29]. Subsequent spatial analyses across Davao City covering 2018 to 2022 documented a strong positive correlation between NDBI and LST (r = 0.6216), a moderate inverse correlation between NDBI and the Normalized Difference Vegetation Index (NDVI, $r = -0.4313$), and an inverse correlation between NDVI and LST (r = −0.2495) [31]. Parallel spatiotemporal assessments confirmed that continuous built-up expansion in Davao City has fragmented vegetative buffers and intensified localized land surface temperatures within dense urban zones [30]. Crucially, all three local investigations evaluated direct spaceborne radiometric LST rather than ground-level air temperature, confirming that NDBI and NDVI provide domain-specific, verifiable explanatory features for modeling surface thermal conditions in Davao City.

To capture the non-linear interactions governing urban microclimates, supervised machine learning techniques have been widely adopted across urban thermal remote sensing [32], [33]. As documented in systematic reviews of environmental spatial data, conventional tree-based ensemble algorithms—most notably Random Forest—readily fit high-dimensional non-linear interactions and threshold effects without requiring parametric distribution assumptions or manual

variable transformations [34]. Random Forest models are capable of generalizing to held-out test datasets when trained on appropriate tabular features; however, standard tabular implementations operate under the core assumption that input observations are independent and identically distributed [34]. A conventional tabular Random Forest evaluates each geographic point as an isolated feature vector, lacking internal architectural mechanisms to execute message passing across irregular spatial topologies or aggregate relational context along contiguous street networks [34]. Consequently, while Random Forest provides a strong non-linear baseline, it cannot natively represent the topological structure of interconnected transportation networks.

A vital methodological distinction must also be drawn between estimating a retrospective annual composite and predicting a forward, held-out annual composite. The vast majority of machine learning studies in the existing literature perform cross-sectional fitting or spatial downscaling, wherein models reconstruct missing pixels or downscale spatial resolutions within the same historical year or observation window [32], [33]. Genuinely sequential annual prediction, wherein a multi-year historical sequence of annual satellite observations (e.g., 2019–2024 annual median composites) is leveraged to forecast the spatial thermal pattern of an unseen future target year (e.g., 2025), represents a substantially more demanding machine learning problem. Under this formulation, the model must generalize across annual climate variations and progressive urban growth rather than interpolating within a known temporal distribution. Furthermore, because satellite observations are aggregated annually to eliminate cloud contamination, the available temporal sequence is inherently short (e.g., $T = 6$ annual steps), which introduces unique optimization challenges that distinguish multi-annual satellite estimation from high-frequency meteorological forecasting.

2.1.6 Evaluation of Spatial Prediction and Road-Connectivity Contribution

Spatial prediction involves using observations from sampled locations to estimate an environmental variable at other geographic locations [69]. Its

evaluation therefore requires attention to spatial dependence, in which observations may remain related across training and evaluation sets [70]. A review of cross-validation strategies explains that random splitting can underestimate predictive error when such dependence is ignored [70]. Consistent with this concern, remote-sensing applications using Random Forest showed that spatial validation revealed prediction limitations obscured by random cross-validation [71], [72]. Together, these findings support evaluating the present study's configurations on common spatially held-out observations, while recognizing that performance estimates apply to the prediction setting assessed rather than automatically extending to other locations or environmental conditions.

Beyond geographic separation, remote-sensing literature cautions that withholding samples from training does not necessarily make them independent of the training data [73]. Research on image classification and tree-species segmentation showed that spatial dependence between training and evaluation samples could produce optimistic assessments of model generalization, [74]. Similarly, an LST study covering five Southeast Asian cities, including Manila, explained that spatial–temporal overlap allowed models to exploit location-specific relationships, distinguishing this evaluation setting from transfer to unseen cities [75]. Applied to road-associated locations, these findings suggest that assigning the same satellite-pixel observation to distinct nodes can create repeated information across partitions if those nodes are separated during splitting [73], [74]. Consequently, evaluation should consider the underlying satellite observations alongside node locations when assessing the independence of held-out data [73], [74], [75].

Accordingly, spatial evaluation should distinguish the assessment of model generalization from the estimation of map-wide accuracy [70]. The literature recommends selecting cross-validation strategies according to the intended prediction task, with spatial blocking offering a way to assess predictions for geographically separated observations [74]. However, spatial separation alone does not guarantee that evaluation statistics provide unbiased estimates of accuracy across an entire mapped area. Following that, a methodological study instead

identifies probability sampling and design-based inference as a basis for estimating map accuracy, highlighting the importance of how evaluation observations are selected [76]. These distinctions support using spatial training, validation, and testing partitions for the present comparison while limiting conclusions to the held-out observations and prediction setting evaluated [70], [76].

Beyond partitioning, graph-learning research emphasizes controlled evaluation when comparing model configurations [77]. GraphSAGE is an inductive graph neural network framework that learns node representations by aggregating features from local neighborhoods [47]. Because neighborhood aggregation enables information exchange among connected nodes, comparing road-connected, spatial-proximity, and self-loops only configurations provides a focused way to examine both the contribution of neighborhood information and the relevance of the chosen connections to LST estimation [47], [65], [77]. Furthermore, Random Forest provides a conventional benchmark supported by its use in LST prediction in Singapore and spatial environmental prediction elsewhere [78], [79]. Using the same historical predictors and held-out observations strengthens the basis for comparing predictive performance, while the comparisons among neural configurations more specifically address relational structure than the comparison between Random Forest and LSTM-GraphSAGE, whose architectures also differ [47], [77].

For the quantitative LST evaluation, root-mean-square error (RMSE) and mean absolute error (MAE) describe prediction-error magnitude in temperature units; RMSE gives greater weight to larger errors through squaring, whereas MAE averages absolute errors [80]. In contrast, the coefficient of determination (R^2) is dimensionless and expresses performance relative to a baseline that assigns the mean observed value, with negative values indicating poorer predictions than that baseline [81]. Together, these measures support examining both error magnitude and relative predictive performance, rather than interpreting either aspect alone [80], [81]. Applied to the present study, their calculation on the same held-out observations provides a common basis for comparing the four configurations.

Finally, the interpretation of derived SUHI maps requires attention to the reference surfaces and observation conditions underlying their temperature contrasts. A synthesis of SUHI research explains that estimated intensity varies with the surface types selected to represent urban and rural environments [82]. Similarly, a comparison of 50 cities identifies spatial scale, topography, vegetation phenology, and observation time as considerations for interpreting SUHI patterns consistently [83]. These findings support interpreting the present study's derived SUHI intensity relative to its defined non-urban reference, while retaining LST prediction as the evaluated task [82], [83]. Furthermore, the distinction between predictive evaluation and map-wide accuracy limits conclusions from held-out LST errors: these errors alone do not establish the accuracy of every mapped location or demonstrate causal effects of road connectivity [76].

Synthesis

The reviewed literature establishes that urban surface temperatures are closely associated with built-up expansion and vegetation loss. Studies in tropical and Philippine cities, including Davao City, consistently identify NDBI and NDVI as useful indicators of surface thermal conditions [14], [22], [29]–[31]. These findings support using Landsat-derived environmental features for annual land surface temperature (LST) estimation, while acknowledging uncertainties associated with cloud cover, retrieval conditions, and mixed pixels [50], [53], [56]. The literature also distinguishes LST from surface urban heat island (SUHI) intensity, which requires comparison with a defined non-urban reference, and from air temperature and human heat exposure [7]–[9]. Thus, LST provides the study's predictive target, while derived SUHI patterns must be interpreted within these measurement boundaries.

Although existing research documents spatial heat clusters and elevated temperatures along transportation corridors, the reviewed literature does not establish whether road-network connectivity provides additional predictive information beyond environmental characteristics and geographic proximity [19],

[25], [31]. Graph-based representations offer a means of investigating this gap by distinguishing connections through roads from relationships based on distance [58], [65]. The proposed study therefore compares road-connected, spatial-proximity, and self-loops-only LSTM–GraphSAGE configurations, alongside a Random Forest benchmark, using consistent historical predictors and held-out observations [47], [77]. Spatially separated evaluation that accounts for shared satellite pixels and temporal overlap, supported by RMSE, MAE, and R^2, provides a basis for assessing this contribution [70]–[75], [80], [81]. Any improvement would demonstrate the predictive usefulness of road topology under the evaluated conditions, without establishing a causal effect on surface warming.

2.2 Related Systems

This section reviews implemented systems relevant to environmental and thermal estimation, focusing on their data inputs, processing methods, spatial representations, outputs, evaluation procedures, and reported limitations. It covers satellite-based LST and SUHI mapping systems, conventional machine-learning thermal estimation systems, raster-based deep-learning thermal mapping systems, graph-based environmental and thermal estimation systems, and spatiotemporal temperature forecasting systems. These implementations are subsequently compared through a summary and gap analysis that examines historical environmental information, road connectivity or geographic proximity, controls for testing graph contribution, spatial dependence and shared satellite observations, and the distinction between LST estimation and SUHI derivation. The section concludes by synthesizing these findings and positioning the proposed system within the reviewed approaches.

2.2.1 Satellite-Based LST and SUHI Mapping Systems

Satellite-based thermal mapping systems differ in whether they use an existing LST product or retrieve LST through a separate processing workflow. The USGS Landsat Collection 2 Level-2 surface-temperature product applies a single-channel algorithm using thermal observations, ASTER emissivity

information and atmospheric auxiliary data [84]. In comparison, [52] implemented a statistical mono-window retrieval in Google Earth Engine using Landsat brightness temperature, water-vapour information and adjusted surface emissivity. MODIS MOD11A2 supplies a different temporal and spatial representation, averaging daily LST observations over eight-day periods on a 1 km grid [85]. These products and implementations provide thermal observations, while SUHI intensity requires an additional calculation involving a defined reference surface.

Philippine implementations illustrate how satellite retrieval is incorporated into local mapping workflows. [86] processed Landsat 8 observations for Quezon City, derived LST through a brightness-temperature and emissivity formulation, and examined thermal variation using OLS and geographically weighted regression. Similarly, [87] obtained Baguio LST from Landsat band 10 through the Project GUHeat QGIS toolbox and calculated the Urban Thermal Field Variance Index (UTFVI). Both workflows characterize temperature relative to the study area's mean through UTFVI, which differs from SUHI intensity defined using a non-urban reference. Vergara et al. further reported that LST decreased between the examined scenes while UTFVI increased, illustrating that the two quantities can indicate different aspects of the thermal pattern. Consequently, absolute LST, UTFVI and non-urban-reference SUHI intensity should retain distinct meanings when comparing systems.

Temporal aggregation also distinguishes the reviewed implementations. [29] used annual mean LST images from Climate Engine to map the occurrence and transitions of thermal clusters in Davao City from 1994 to 2019. [23], in contrast, examined yearly maximum-based Landsat thermal maps for Manila City and combined spatial statistics with suitability analysis to identify intra-urban action classes. The Baguio analysis used two individual March scenes, whereas [88] processed dry-season Landsat collections and fitted separate LST models for three years in Da Nang. These outputs represent different temporal statistics and observation windows, so their differences should not be interpreted as equivalent measures of annual thermal conditions. Hoang et al. also provided inconsistent

descriptions of the thermal-processing method, requiring clarification before their LST data can be treated as equivalent to the USGS surface-temperature product.

For explicit SUHI intensity mapping, [89] implemented the Simplified Urban-Extent algorithm in Google Earth Engine. The system calculates the difference between mean urban LST and mean non-urban, non-water LST within an urban extent, with additional filtering related to elevation and urban coverage. The authors nevertheless identified limitations associated with older urban boundaries, uneven reference geometry and clear-sky sampling. Reference design is also central to the Adaptive Synchronous Extraction method of [90], whose published abstract describes adaptive background-reference selection and reports that topography, missing LST observations and surrounding urban areas can affect the estimated intensity. These systems indicate that the reference surface forms part of the measurement definition rather than serving only as a final mapping adjustment.

Satellite-based mapping remains constrained by retrieval quality and the spatial and temporal support of available observations. The USGS documentation identifies surface-temperature gaps associated with missing ASTER emissivity data, errors near clouds and cloud shadows, and problems associated with emissivity adjustment [84]. Moreover, [23] distinguish the 100 m native Landsat thermal resolution from the 30 m delivered analysis grid. Assigning such raster values to multiple road-associated locations therefore does not create additional independent thermal measurements. Likewise, an annual median composite summarizes the available valid observations and does not preserve the seasonal or diurnal variation examined by systems with finer temporal sampling. These considerations should accompany comparisons of mapped outputs and subsequent predictive estimates.

The proposed system differs from these mapping workflows by using annual median composites of Landsat 8 environmental observations as historical inputs to an LSTM-GraphSAGE model at road-associated locations. Its prediction

target is LST, after which SUHI intensity is derived using the median of valid annual-composite LST pixels within a same-year non-urban reference mask. Road-connected adjacency is examined as a candidate relational representation rather than assumed to describe physical heat transfer. The system therefore extends the mapping workflow with a controlled predictive comparison, while retaining the retrieval uncertainty and observational-support constraints of its satellite inputs.

2.2.2 Conventional Machine-Learning Thermal Estimation Systems

Conventional machine-learning thermal estimation systems in the urban climate literature have historically evolved from parametric regression models to non-parametric, tree-based ensemble algorithms [32], [33]. Traditional approaches to thermal mapping relied predominantly on ordinary least squares Multiple Linear Regression (MLR) and geostatistical interpolation methods such as ordinary kriging or regression kriging [33], [34]. These parametric systems presuppose linear, additive relationships between explanatory spatial predictors (such as vegetation indices, albedo, and building density) and the observed thermal response. However, urban surface and atmospheric microclimates are governed by highly coupled, non-linear radiative and turbulent heat transfer processes that violate standard linear and homoscedastic assumptions [32], [35]. Consequently, linear models frequently exhibit severe underfitting when confronted with complex micro-urban variations, displaying extreme sensitivity to multicollinearity among correlated surface descriptors and susceptibility to spatial outliers [35].

To overcome the architectural rigidity of linear formulations, tree-based ensemble algorithms, most notably Random Forest (RF) regression, have become the benchmark conventional machine-learning methodology for urban heat island and thermal estimation systems [32], [33]. By recursively partitioning the multidimensional feature space through bootstrapped bagging and randomized feature subsets at each decision split, Random Forest natively captures high-order non-linearities and threshold effects without requiring prior data transformation or rigid parametric assumptions. In a micro-urban comparison conducted in Londrina,

Brazil, researchers evaluated daytime and nighttime MLR against Random Forest for fine-scale air temperature (T_{air}) modeling at 10-meter spatial and 1-hour temporal resolutions [35]. Leveraging mobile traverse measurements alongside a comprehensive suite of land cover, urban geometry, and atmospheric predictors, their findings revealed that Random Forest substantially outperformed MLR, explaining over 96% of the thermal variance across daytime and nighttime conditions ($R^2 > 0.96$), whereas MLR achieved modest explanatory power during the day ($R^2 = 0.64$) and deteriorated severely at night ($R^2 = 0.34$) [35]. Interpretability analysis using SHAP (Shapley Additive exPlanations) values confirmed that meteorological variables and surface land cover were the dominant predictors, and that RF effectively resolved the complex interactions that caused linear models to fail under distinct synoptic weather regimes [35]. However, because this investigation modeled near-surface mobile air temperature (T_{air}) rather than spaceborne radiometric LST, its numerical metrics cannot be interpreted as proof of satellite LST estimation performance, serving instead as adjacent evidence of Random Forest's superior non-linear fitting capacity in micro-urban climate modeling.

In the Philippine context, localized conventional machine-learning applications have similarly demonstrated the necessity of non-linear estimation frameworks for satellite-derived LST. Investigating Davao City's urban thermal environment, researchers tested Multiple Regression Models (MRM) and Quantile Regression Models (QRM) to evaluate how environmental and anthropogenic factors—including NDBI, Enhanced Vegetation Index (EVI), albedo, elevation, and population density—influenced Landsat 8-derived LST across the 1994–2019 period [29]. While standard multiple regression provided an undifferentiated average relationship, quantile regression revealed that the explanatory significance of individual predictors varied dynamically across the LST distribution: albedo was the primary explanatory variable at the lowest thermal quantile (10th percentile), elevation dominated at the 25th percentile, whereas NDBI was the only statistically significant predictor across the median and upper quantiles (50th, 75th, and 90th percentiles) [29]. Similarly, in northern Mindanao, non-linear land-cover

transformations were empirically mapped to evaluate multi-decadal LST shifts across river basin catchments [24]. These empirical results demonstrate that urban thermal dynamics in Mindanao exhibit distinct non-linear regime shifts across different land-cover categories and thermal percentiles.

Despite its proven empirical strength in fitting tabular datasets, conventional Random Forest regression operates under fundamental architectural constraints when deployed in spatially continuous, networked urban environments. As highlighted in systematic reviews of environmental machine learning, standard tabular implementations of Random Forest treat training observations as independent and identically distributed instances [34]. A standard tabular model processes each geographic observation as an isolated feature vector, lacking internal mechanisms to account for spatial distance decay, spatial lag, or directional connectivity between adjacent sites unless such relationships are manually engineered into static spatial covariates [34], [40]. When applied to satellite remote sensing pixels or road-network nodes, conventional tabular models fail to capture the continuous spatial field of the urban fabric: they cannot dynamically aggregate information across neighboring urban street canyons or recognize that contiguous paved corridors share interconnected thermal microclimates. Importantly, this structural property does not imply that Random Forest is incapable of generalizing to held-out test data; rather, it establishes that Random Forest cannot perform topological message passing across graph edges. Random Forest therefore represents a highly competitive, academically rigorous baseline against which graph neural network architectures must be objectively evaluated.

2.2.3 Raster-Based Deep-Learning Thermal Mapping Systems

In response to the feature-engineering bottlenecks of tabular machine learning, raster-based deep-learning systems have emerged as a prominent methodology for urban thermal mapping and surface heat island detection. Unlike point-based tabular models, raster deep-learning frameworks process satellite imagery and geospatial layers directly as multi-channel, continuous two-dimensional pixel arrays [32], [33]. By utilizing Convolutional Neural

Networks (CNNs) and hierarchical encoder-decoder segmentation architectures, these systems automatically extract spatially correlated patterns across multi-pixel receptive fields. Convolutional filters slide across the image array, computing localized inner products that capture fine-grained spatial textures, gradient boundaries, and contextual land-cover arrangements without requiring manual spatial feature extraction [36], [45].

A prominent empirical implementation of raster-based deep learning for thermal mapping is the open-source geospatial framework developed for urban heat island detection in Lagos, Nigeria [36]. Addressing the scalability and resolution limitations of classical empirical indices and statistical regression, the authors constructed an end-to-end convolutional pipeline based on a modified U-Net architecture integrated with spatial attention mechanisms. The system ingested gridded rasters comprising multispectral vegetation indices (NDVI), impervious surface area percentages, building footprint density, and categorical land-use classifications to simultaneously estimate continuous LST and delineate discrete UHI hotspot boundaries across the rapidly urbanizing tropical megacity [36]. The incorporation of attention gates within the U-Net skip connections allowed the network to suppress irrelevant background spectral noise while highlighting critical transitional boundaries between densely built impervious cores and cooling water bodies, achieving strong empirical alignment with ground-level urban morphology [36]. In a complementary urban analytics application, deep learning models were deployed for multi-domain urban planning in Krasnodar, utilizing CNNs for land-use classification (91.6% accuracy) and U-Net models for building footprint segmentation (94.0% accuracy) from satellite imagery integrated with OpenStreetMap vector layers [45]. These implementations illustrate the capability of convolutional architectures to extract rich, contextual spatial representations from raw geospatial rasters.

Notwithstanding their notable successes in image segmentation and spatial pattern recognition, raster-based deep-learning systems encounter substantial representation differences when applied to linear, irregularly arranged urban infrastructure. As established in graph signal processing theory, standard

convolutional neural networks are fundamentally confined to regular Euclidean domains—such as uniform rectangular grids where each internal pixel possesses an identical, fixed neighborhood geometry (e.g., eight cardinal and diagonal neighbors) [39] In contrast, the built urban environment is structured along irregular, non-Euclidean street networks characterized by variable node degrees, asymmetric segment lengths, and non-uniform connectivity [37], [61]. When a raster CNN applies a rigid square convolutional kernel (e.g., 3 × 3 or 5 × 5 pixels) over an urban scene, it pools all pixels within that rectangular window, aggregating spectral radiance across natural barriers, physical walls, private building interiors, and disjointed street segments without regard to functional infrastructure connectivity [37].

This architectural distinction represents a fundamental difference in spatial representation rather than proof of universal superiority. For continuous, space-filling phenomena across broad geographic landscapes, raster CNNs provide an effective, mathematically natural modeling framework. Conversely, when the research objective focuses specifically on transportation corridors, road intersections, and street-associated infrastructure, graph-based representations allow models to isolate road-associated nodes and propagate relational information strictly along connected transit corridors [38], [39]. Furthermore, existing raster deep-learning thermal systems are primarily designed for static, point-in-time image segmentation or cross-sectional spatial detection rather than long-term temporal sequence modeling [33], [36]. In tropical urban environments like Davao City, obtaining cloud-free, spatially contiguous raster image scenes across consecutive years is severely impeded by persistent monsoonal cloud cover, making uninterrupted 2D image-to-image sequence modeling difficult to execute reliably [7]. Acknowledging the relatively thin body of published literature directly applying deep convolutional architectures to multi-temporal thermal infrared remote sensing reinforces the value of exploring graph-structured representations for road-associated thermal modeling.

2.2.4 Graph-Based Environmental and Thermal Estimation Systems

Graph-based environmental estimation systems represent observation sites or spatial units as nodes and their relationships as edges. Graph neural networks then aggregate information from connected nodes to estimate environmental variables. However, the meaning of a connection depends on how the graph is constructed. The reviewed systems employ geographical proximity, similarity between urban characteristics, and spatiotemporal adjacency. These representations provide relevant examples for examining how relational information contributes to thermal estimation, although their results do not establish the usefulness of road connectivity for the proposed study [40]

For direct thermal estimation, [41]developed a graph convolutional network to estimate annual daytime and nighttime surface urban heat island (SUHI) intensity across spatial subunits in Hong Kong. The spatial subunits formed the nodes, while natural and built-environment characteristics served as node features. An undirected K-nearest-neighbour graph with self-loops represented relationships between the subunits. The model was evaluated against Random Forest, XGBoost, and multilayer perceptron baselines. Furthermore, the study compared proximity-based and contiguity-based graphs and examined sensitivity to neighbour count and network depth. These experiments illustrate the evaluation of alternative graph representations. Nevertheless, the reported selection of graph settings using test accuracy warrants caution when interpreting the resulting performance. The authors also acknowledged limitations associated with older thermal data, inconsistent observation years, coarse thermal resolution, and application to a single city. Compared with this framework, the proposed system uses road-associated locations as nodes and estimates LST before deriving SUHI.

Another direct LST application is the X-STGNN framework introduced by [91] for Chennai, Hyderabad, and Delhi. The framework combines graph convolution, LSTM learning, feature fusion, and temporal attention using remote-sensing indicators, Local Climate Zone information, urban morphology,

and meteorological variables. The published abstract reports component ablations and leave-one-city-out evaluation. This implementation supports the relevance of combining temporal and graph learning for LST prediction. However, the accessed abstract does not specify the exact node definitions, edge-construction rules, or prediction horizon. Its correspondence with a road-connected graph therefore cannot be established from the available information.

For a related thermal variable, [92] implemented a GraphSAGE framework for street-level air-temperature estimation in Chicago. Sensor locations formed the nodes, and geographical distance together with land-use similarity determined their connections. Meteorological information from WRF simulations and urban geographical features were processed through a fully connected encoder, GraphSAGE layers, and a fully connected decoder. Unlike an estimation design using only preceding observations, its meteorological input window was centred on the target hour and included subsequent hours. Evaluation included a Gaussian-process regression benchmark, geographical-feature ablations, and leave-one-sensor-out experiments. The latter examined estimation at sensor locations excluded during training. Nevertheless, location holdout alone does not establish independence of shared meteorological and geographical inputs. The system differs from the proposed study in its air-temperature target, sensor-based nodes, and distance-and-similarity graph.

More closely related to the proposed model architecture, [93] developed the Mix-n-Scale framework for six-hour air-temperature forecasting at 16 fixed observation sites in Hong Kong. Historical observations were encoded through LSTM and subsequently processed using GraphSAGE or graph attention layers. GraphSAGE used an undirected graph with mean aggregation, whereas the attention-based model used directed relationships. Training, validation, and testing used observations from 2016–2019, 2020, and 2021, respectively. The experiments varied neighbour selection, neighbour count, and graph depth. Although an LSTM benchmark supplied a non-graph comparison, its architecture differed from the graph models, and the main text did not report a self-loops-only GNN result.

Consequently, the comparison does not isolate adjacency from architectural differences. The authors also clarified that the task concerned forecasting at fixed observation sites rather than spatial prediction at unmeasured locations.

Beyond thermal variables, [94] implemented GC-LSTM for PM2.5 forecasting. Monitoring stations formed the nodes, with inverse-distance-weighted connections between stations separated by less than 200 km. Graph convolution processed spatial information before LSTM learned temporal relationships from historical observations. The model was compared with multiple linear regression, a feedforward neural network, and LSTM using randomly assigned temporal blocks. This system demonstrates temporal–graph integration for an environmental variable, although its processing order and station-proximity graph differ from the proposed LSTM-GraphSAGE design. The authors reported concentration-dependent prediction bias and difficulty estimating abrupt extreme values.

In contrast to forecasting, [40] developed GNN-STI for interpolation at unsampled locations. Their approach uses a spatiotemporal Voronoi-adjacency graph and a locally stationary diffusion kernel to represent dependence among observations. Evaluation included simulated data and PM2.5 observations across China, covering both sample-rich and sample-poor areas. The accessed abstract establishes the interpolation objective and graph family but does not provide sufficient detail to compare node features, split procedures, or graph ablations. Its relevance therefore lies in demonstrating another implemented relational representation for environmental estimation.

Taken together, these systems show that graph learning can be implemented using several definitions of spatial and temporal relationships. Their evaluations also address different questions: alternative-graph experiments examine graph construction, temporal holdout examines performance on later observations, and sensor holdout examines transfer to excluded locations. These procedures should therefore be interpreted according to their respective purposes. Building on these

distinctions, the proposed study examines road-connected adjacency as a candidate relational representation and compares it with self-loops only and a spatial-proximity graph. Any observed performance difference would indicate predictive usefulness under the study conditions; it would not establish a causal relationship or identify roads as physical heat-transfer pathways.

2.2.5 Spatiotemporal Temperature Forecasting Systems

Spatiotemporal temperature forecasting systems require modeling architectures capable of capturing both temporal sequential dynamics and non-Euclidean spatial dependencies. In the temporal domain, recurrent neural networks, most prominently the Long Short-Term Memory (LSTM) architecture, provide a mathematically rigorous solution to the vanishing and exploding gradient problems that hinder standard recurrent networks over long sequence lengths [46]. Through the coordinated operation of multiplicative input, forget, and output gates operating on an internal constant error carousel, LSTMs preserve, update, and discard historical feature representations across arbitrary time horizons [46]. In urban computing and spatial time series forecasting, LSTMs have proven highly effective for modeling sequential dependencies across complex urban infrastructure, successfully extracting latent temporal trajectories from historical environmental observations [45], [95].

In the spatial domain, Graph Neural Networks (GNNs) generalize deep learning operations to irregular, non-Euclidean topologies by representing physical or conceptual systems as graphs $G = (V, E)$, where vertices V denote observation entities and edges E encode relational interactions [39]. In their foundational theoretical analysis, Ruiz, Gama, and Ribeiro established that graph convolutional architectures possess mathematical equivariance to node permutations and structural stability to graph deformations [39]. These stability properties provide a mathematical guarantee that GNNs remain robust against minor perturbations in network topology, mathematically validating the transferability of learned graph filters across varying network sizes and geometries [39]. Extending graph learning to large, dynamically evolving datasets, Hamilton, Ying, and Leskovec formulated

GraphSAGE (Graph SAmple and aggreGatE), an inductive representation learning framework [47]. Unlike transductive spectral graph convolutions that require the entire Laplacian matrix to be present during optimization, GraphSAGE learns parameterized aggregator functions—such as mean, pooling, or LSTM aggregators—that sample and aggregate feature vectors from localized node neighborhoods [47]. This inductive formulation enables the trained network to generalize seamlessly to newly added nodes, altered subgraphs, or entirely unseen geographic sectors without requiring global retraining, making GraphSAGE exceptionally well-suited for municipal infrastructure graphs [47] Spatiotemporal Graph Neural Networks (STGNNs), which integrate temporal recurrent or convolutional layers with spatial graph convolutions, have consequently seen extensive deployment across environmental and thermal forecasting domains, as synthesized in comprehensive reviews of 366 STGNN systems [95] and broader urban computing surveys [42]. Within the specific realm of micro-urban and transportation temperature modeling, an empirical study deployed a high-resolution GNN framework to forecast Road Surface Temperature (RST) along road networks in Sapporo, Japan [43]. By discretizing road corridors into ~100-meter intervals and combining graph spatial modeling with telemetry from road weather observatories and atmospheric mesh forecasts, their system achieved root mean square errors below 2.0 °C for future horizons up to 24 hours, substantially outperforming operational municipal maintenance baselines [43].

In environmental spatial interpolation, an advanced graph interpolation architecture utilized spatiotemporal Voronoi-adjacency graphs and locally stationary diffusion kernels to interpolate continuous environmental fields across irregular monitoring stations in China, demonstrating significant accuracy gains over classical geostatistical and machine learning baselines across densely sampled and sparse spatial domains [40]. Furthermore, researchers applied a Graph Convolutional Network to analyze diurnal SUHI variations in hyper-dense Hong Kong, integrating built and natural urban features into district-level graphs to achieve test R^2 values of 0.9173 for daytime and 0.8789 for nighttime SUHI, while identifying non-linear interactions between vegetation cooling and dense building

clusters through integrated gradient interpretability [41]. Complementary meteorological STGNN implementations, such as the variable-resolution unstructured graph neural network developed for very short-range near-surface weather forecasting in Guangdong [96] and the Spatial-Temporal Synchronous Graph Convolutional Network (STSGCN) formulated for localized network forecasting [97], further demonstrate the architectural capacity of graph convolutions to resolve complex environmental correlations.

However, a vital methodological distinction must be recognized between operational meteorological forecasting and multi-annual satellite LST estimation. The forecasting systems documented in the literature [43], [96], [97] are explicitly designed for short-term dynamic forecasting across hourly, diurnal, or multi-day lead times. These short-range systems ingest high-frequency telemetry feeds, atmospheric reanalyses, or real-time traffic loops, capturing transient thermodynamic variations driven by immediate weather fronts, turbulent boundary layer shifts, and diurnal solar angles. In sharp contrast, the problem addressed in annual satellite LST estimation is fundamentally climatological and structural: the objective is to predict the annual median radiometric surface temperature of urban road nodes in a future target year (e.g., 2025) using a continuous multi-year historical sequence (e.g., 2019–2024) of cloud-screened satellite observations. At an annual temporal aggregation, high-frequency atmospheric fluctuations and synoptic weather noise are filtered out, leaving the target temperature governed by multi-year urban morphology, cumulative impervious expansion, and vegetative land-cover transitions [7], [33].

This temporal framing introduces specific learning challenges and dictates rigorous validation standards. In annual estimation, the temporal sequence length is inherently short (e.g., $T = 6$ annual steps), which creates a high risk of overfitting in recurrent neural architectures if model capacity is not properly constrained. Learning meaningful temporal representations from short sequences requires regularized architectures with strong structural inductive biases. Furthermore, the evaluation protocol must employ strictly chronological, held-out-year validation rather than randomized k-fold cross-validation. Randomly partitioning time-series

observations across training and validation splits induces severe temporal data leakage, as future macroeconomic and climatological trends bleed into past training steps, producing artificially inflated accuracy metrics. Only forward, held-out-year validation (training on historical years 2019–2024 to predict unseen target year 2025) provides an honest, academically defensible test of whether spatiotemporal graph neural networks generalize across evolving urban thermal landscapes.

2.2.6 Summary and Gap Analysis of Related Systems

Table 2.X compares selected implemented systems discussed in Sections 2.2.1–2.2.5 according to historical environmental information, road connectivity or geographic proximity, controls for testing graph contribution, spatial dependence and shared satellite observations, and the distinction between LST estimation and SUHI derivation. The comparison distinguishes the availability of observations from the use of a temporal learning component, and geographic predictors from relationships explicitly represented through graph edges. Entries marked “Not reported” identify details absent from the supplied section descriptions; they do not establish that the original studies omitted those procedures.

The systems differ in their temporal inputs and spatial representations. [29] used annual mean LST images to examine thermal patterns in Davao City, whereas [93] combined historical station observations with LSTM and graph layers for short-term air-temperature forecasting. [92] represented relationships among sensors using distance and land-use similarity, while [36] used raster neighborhoods in a U-Net framework. [41] compared proximity-based and contiguity-based graph constructions, providing an example of evaluating alternative spatial relationships. [43] forecast road-surface temperature at locations along roads, although the supplied description does not establish the graph’s edge-construction rule.

The resulting gap concerns the combination of controlled adjacency comparisons and spatial evaluation for annual LST estimation at road-associated locations. Shao et al. [41] examined alternative graphs, and [93] varied neighbor

selection, neighbor count, and graph depth. However, [93] non-graph benchmark used a different architecture, and a self-loop-only GNN result was not reported in the main-text description. The supplied sections do not document the complete comparison of road-connected, self-loops only, spatial-proximity graph, and Random Forest configurations using the same historical predictors and held-out observations. Reporting of shared satellite pixels is also incomplete across the selected satellite-based systems; this reporting gap does not demonstrate that leakage occurred. Moreover [87] used an area-mean-based UTFVI, whereas [89] calculated an urban-minus-non-urban LST difference, requiring these outputs to retain distinct definitions.

The proposed system addresses these comparison dimensions through LSTM-GraphSAGE configurations using road-connected adjacency, self-loops only, and a spatial-proximity graph, together with a Random Forest benchmark evaluated on the same held-out observations. Spatial partitioning and controls for shared satellite pixels support evaluation across road-associated locations while accounting for the observational support of the inputs. LST remains the estimation target, and SUHI intensity is derived afterward using a defined non-urban reference. Road connectivity is therefore evaluated as a candidate relational representation, without assuming improved performance or interpreting road edges as physical heat-transfer pathways. Chapter III presents the procedures for implementing and evaluating these comparisons.

Table 2.1 Comparative summary of selected related systems

System/author/year	Purpose and model	Historical environmental information	Road connectivity or geographic proximity	Controls for testing graph contribution	Spatial dependence and shared satellite pixels	LST estimation and SUHI derivation

Tinoy et al. (2019) [29]	Davao thermal mapping and regression analysis.	Annual mean LST, 1994–2019.	Not reported in 2.2.1–2.2.2.	Not applicable: non-graph system.	Thermal clusters examined; shared-pixel controls not reported in 2.2.1–2.2.2.	LST mapped and modeled; non-urban-reference SUHI not reported in 2.2.1–2.2.2.
Vergara et al. (2023) [52]	Baguio LST retrieval and UTFVI mapping.	Two individual March scenes.	Not reported in 2.2.1.	Not applicable: non-graph workflow.	Split and shared-pixel controls not reported in 2.2.1.	GUHeat-derived LST; UTFVI uses the study-area mean.
Chakraborty and Lee (2019) [54]	Simplified Urban-Extent algorithm for SUHI mapping.	Observation period not reported in 2.2.1.	Urban-extent masks and elevation filtering.	Not applicable: non-graph algorithm.	Clear-sky sampling limitations; shared-pixel validation controls not reported in 2.2.1.	Mean urban LST minus mean non-urban, non-water LST.
Oukawa et al. (2022) [35]	Air-temperature modeling using Random Forest and multiple linear regression.	Historical input sequence not reported in 2.2.2.	Land-cover and geometry predictors; explicit relational structure not reported in 2.2.2.	Not applicable: non-graph models.	Spatial split not reported in 2.2.2; satellite thermal pixels not applicable to the air-temperature target.	Not applicable: air temperature is the target.

Akintola and Neziri (2025) [36]	LST and hotspot mapping using modified U-Net.	Static cross-sectional inputs described.	Raster neighborhoods; road edges not reported in 2.2.3.	Not applicable: non-graph model.	Split and shared-pixel controls not reported in 2.2.3.	Continuous LST and hotspots; non-urban-reference SUHI not reported in 2.2.3.
Shao et al. (2026) [41]	Annual daytime and nighttime SUHI estimation using GCN.	Sequence learning not reported in 2.2.4–2.2.5.	Spatial subunits; proximity and contiguity graphs.	Alternative graphs, neighbor counts, depth variation, and non-graph baselines.	Split and shared-pixel controls not reported in 2.2.4–2.2.5.	SUHI estimated directly; reference definition not reported in 2.2.4–2.2.5.
Yu et al. (2024) [56]	Street-level air-temperature estimation using GraphSAGE.	Centered temporal window includes future-hour observations.	Sensor graph based on distance and land-use similarity.	Feature ablations and GPR benchmark; edge-only control not reported in 2.2.4.	Leave-one-sensor-out evaluation; shared-data independence unestablished; satellite thermal pixels not applicable.	Not applicable: air temperature is the target.
Wang et al. (2026) [57]	Six-hour air-temperature forecasting: LSTM with GraphSAGE or GAT.	Station histories; 2016–2019 training, 2020 validation, 2021 testing.	Station graphs varying neighbor selection and count.	Graph variations; LSTM benchmark with different architecture; self-loop-only main-text result not reported.	Temporal partition at fixed sites; satellite thermal pixels not applicable.	Not applicable: air temperature is the target.

Takedomi et al. (2025) [43]	Road-surface temperature forecasting using GNN.	Road observations and weather forecasts; historical input window not reported in 2.2.5.	Road-associated locations; edge-construction rule not reported in 2.2.5.	Not reported in 2.2.5.	Spatial split and dependence controls not reported in 2.2.5; satellite thermal pixels not applicable.	Not applicable: road-surface temperature is the target.
Proposed system	Annual LST estimation using LSTM-GraphSAGE; Random Forest benchmark.	Historical annual LST, NDVI, and NDBI.	Road-connected, self-loops only, and spatial-proximity graph configurations.	Four configurations evaluated on the same held-out observations.	Spatial partitioning and controls for shared satellite pixels.	LST estimated first; SUHI derived afterward using a non-urban reference.

2.2.7 Synthesis and Positioning of the Proposed System

The reviewed systems provide complementary approaches to representing and estimating urban thermal conditions. Philippine satellite-based implementations establish relevant local applications while illustrating differences in temporal aggregation and thermal indicators [29], [87]. Conventional regression and Random Forest systems estimate temperature from explanatory variables, whereas raster-based deep-learning systems incorporate spatial context through image neighborhoods [35], [36]. Graph-based implementations introduce relationships among observation locations or spatial units, and some evaluate alternative graph constructions or combine temporal encoding with neighborhood aggregation [41], [92], [93]. However, their targets, observation frequencies, graph definitions, and evaluation procedures differ. Accordingly, results for air temperature, road-surface temperature, LST, and SUHI should be interpreted

within their respective measurement definitions and study conditions. Collectively, these implementations support investigating temporal and relational modeling, while leaving the predictive contribution of road-connected adjacency to annual LST estimation unresolved within the reviewed comparison.

Building on these findings, the proposed system combines historical environmental encoding and neighborhood aggregation through LSTM-GraphSAGE to estimate annual LST at road-associated locations in Davao City. Its contribution centers on evaluating road-connected adjacency against self-loops only and a spatial-proximity graph, alongside Random Forest using the same historical predictors and held-out observations. Spatial evaluation and controls for shared satellite pixels address the dependence associated with assigning raster observations to multiple locations. Furthermore, LST remains the prediction target, while SUHI intensity is derived afterward using a defined non-urban reference. This design positions road connectivity as a candidate relational representation whose predictive usefulness must be tested, without implying causality or treating road edges as physical heat-transfer pathways. The resulting comparison is intended to clarify whether road-connected information contributes to annual LST estimation under the study conditions and to support subsequent SUHI mapping.

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